Luiza Lober de Souza Piva

PhD candidate | Physics

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Plataforma Lattes

São Carlos, SP - Brasil

Personal website

LinkedIn

Education

PhD in physics

University of São Paulo (USP)

Complex networks dynamics through the use of machine learning, focusing on epidemic modeling and synchronization.

Project title: Dynamics of nonlinear systems using machine learning

Funding: FAPESP

2022 – ongoing

Extension: Complex data mining

State University of Campinas (UNICAMP)

Five month course on data science and machine learning tools and its applications.

Final project: Team solution to the RSNA-MICCAI Brain tumor radiogenomic classification competition at Kaggle

2022

MSc. physics

State University of Campinas (UNICAMP)

Particle physics - Phenomenology in heavy-ion collisions.

Project: Study of initial conditions using PYTHIA to hydrodynamic simulations

Funding: CAPES

2020 – 2022

BSc. physics

State University of Campinas (UNICAMP)

2016 - 2019

Associated undergrad research projects:

1. Study of elliptic flow and simulation techniques in ultrarelativistic heavy-ion collisions (**Funding**: FAPESP). 2018-2019;

2. Development of a light detector to a LT-STM (**Funding**: FAPESP). 2017-2018.

Work experience

Teaching assistant (PED)

State University of Campinas (UNICAMP)

August 2021 - December 2021: "Física Experimental III" (F 329), class A;

March 2021 - July 2021: "Física Experimental III" (F 329), class T.

Undergraduate teaching assistant (PAD)

State University of Campinas (UNICAMP)

August 2019 - December 2019: "Física Geral IV" (F 428), class A.

Quanta Jr.

State University of Campinas (UNICAMP)

Junior business with educational and modeling solutions developed by physics undergrads.

December 2018 - April 2019: Vice president;

August 2018 - December 2018: Advisor;

March 2018 - August 2018: Trainee.

Technical knowledge

Programming languages: Python, C, C++, HTML5.
Machine Learning Knowledge: Sparse regression algorithms, Random Forests and similar regression models, time series forecast, symbolic regression and PINNs, GCNs.
Code packages: ROOT, LaTeX, Microsoft Office, Libre Office, Jupyter Notebook, Git, SSH, Matplotlib, Numpy, Scipy, Scikit Learn, TensorFlow and Pandas.

Events and presentations

Conference on Complex Systems (2023)

Presentation title: Extracting the governing equations of epidemics through sparse regression

Oct. 2023

Ada Lovelace Day - Women in science congress

Presentation title: [pt-br] Extraindo as equações de sincronização em redes complexas através de regressão esparsa

🏆 2nd place with the oral presentation "Extraindo as equações de sincronização em redes complexas através de regressão esparsa" on Ada Lovelace Day.

📅 Nov. 2022

Applications of Nonlinear Systems to Socio-Economic Complexity

Presentation title: Applying a sparse regression model in complex networks

📅 Oct. 2022

XLIV Brazilian Meeting on Nuclear Physics

Presentation title: Initial conditions using PYTHIA for hydrodynamic simulations

📅 Nov. 2021

The VIth Initial Stages

Presentation title: Using PYTHIA as an initial condition generator for hydrodynamics

📅 Jan. 2021

XXVII PIBIC Congress

Presentation title: Elliptic flow in ultrarelativistic heavy-ion collisions

🏆 Honorable mention to the poster "Elliptic flow in ultrarelativistic heavy-ion collisions" (2019)

📅 Sep. 2019

XL National Congress of Particle Physics and Fields

Presentation title: Elliptic flow in ultrarelativistic heavy-ion collisions

📅 Sep. 2019

XXVI PIBIC Congress

Presentation title: Development of a light detector for a Low Temperature Scanning Tunneling Microscope (LT-STM)

📅 Oct. 2018

Awards

2nd place to a oral presentation

The presented work, titled "Extraindo as equações de sincronização em redes complexas através de regressão esparsa", was presented at "Seminário Mulheres na Ciência" that took place in State University of São Paulo (USP São Carlos).

Honorable mention – Projeto de iniciação científica

To the poster presented at "XXVII Congresso PIBIC" that took place in Universidade Estadual de Campinas (UNICAMP).

Languages

Portuguese

Native speaker

English

Professional proficiency.